

OPTIMAL WITHDRAWALS IN A DIFFUSION MODEL WITH STATE-DEPENDENT CONTROL RATES*

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Abstract. We consider a classical stochastic control problem in which a diffusion process is controlled by a withdrawal process up to a termination time. The objective is to maximize the expected discounted value of the withdrawals until the first-passage time below level zero. In this work, we are considering absolutely continuous control strategies in a general diffusion model. Our main contribution is a solution to the control problem under study, which is achieved by using a *probabilistic* guess-and-verify approach. We prove that the optimal strategy belongs to the family of bang-bang strategies, i.e., strategies in which, above an optimal barrier level, we withdraw at the highest-allowed rate, while no withdrawals are made below this barrier. Some nontrivial examples are studied numerically.

Key words. stochastic control, absolutely continuous strategies, diffusion model, refraction strategies

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1. Introduction. We consider a classical stochastic control problem in which a nonnegative and increasing control process, called hereafter a *withdrawal process*, is subtracted from a given diffusion process. The objective is to maximize the expected discounted value of the withdrawals made until the termination time, chosen here as the first-passage time (of the controlled process) below level zero. The challenge in this control problem lies in finding an optimal balance between large amounts of withdrawals and longevity of the process. This problem has interpretations in areas such as storage and inventory modelling [18], in insurance and financial mathematics [11, 15, 19], and also in harvesting of populations [3]; for example, in the latter two interpretations, a withdrawal strategy is often called a dividend policy and a harvesting plan, respectively. In the existing literature, various models, admissible withdrawal strategies, and performance functions have been studied. Without transaction costs, the problem typically results in a singular control problem [18], while with fixed transaction costs imposed on each withdrawal, it becomes an impulse control problem [15, 4].

In this work, we consider absolutely continuous controls in a general diffusion model. The novelty of this control problem lies in restricting the control rate process to be less than a level-dependent bound, i.e., a *bound function* applied to the current state of the controlled process, in the spirit of [3], [16] and [13]. In their paper,

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the authors of [3] have considered two versions of this problem in a logistic diffusion model, respectively with a constant bound function and a linear bound function. Similarly, [16] and [13] have considered a simple Brownian motion model with a linear bound function and with a nondecreasing and concave bound function, respectively. Previously, [19] had considered a general diffusion model but with a constant bound function. Consequently, our paper generalizes these works by tackling the problem in a general diffusion model with a nondecreasing and concave bound function.

Our main contribution is a solution to the control problem described above, which is achieved by using a *probabilistic* guess-and-verify approach. This method has the advantage of identifying explicitly the dynamics of an optimal strategy and providing an analytical expression for the value function of the control problem. Indeed, we compute first the performance function of an arbitrary refraction strategy, i.e., a bang-bang strategy in which withdrawals are made at the highest possible level-dependent rate above a given threshold and no withdrawals are made below the threshold. Using this semi-explicit analytical form, we then identify the (candidate) optimal threshold level; this is the most challenging step. The final step is the verification procedure, which is a confirmation that this candidate optimal refraction strategy is in fact optimal within the set of all admissible withdrawal strategies.

Another contribution worth mentioning is the analytical study of a functional (I_F below) needed to identify the optimal barrier level. This function is given by the application of the resolvent operator of the underlying diffusion (without killing) to the bound function (F below). In [16], this expectation was computed explicitly, thanks to the combined simplicity of the Brownian model and the linear bound function. In [13], also in a Brownian model but with a general nondecreasing and concave bound function, in which case no explicit expression for this functional is expected, a significant amount of effort was devoted to proving that it is a concave function. Our approach, in a general diffusion model, is based on the notion of concave envelope and on classical comparison results for solutions of ordinary differential equations (ODEs). As in [13], we prove that this function is concave, which means in particular that the resolvent operator preserves concavity. As we will show in section 5, concavity of the functional I_F , as well as the concavity and convexity properties of the fundamental solutions of the infinitesimal generator, will be crucial to identifying the (candidate) optimal threshold. Previously, in [5] the authors studied a singular control problem with running costs driven by a general diffusion; in contrast to our approach, the authors used the property of *concavity with respect to a function* to prove the existence of a candidate optimal threshold in their context.

Finally, we illustrate our results for some examples of the underlying diffusion model and of the bound function. In particular, when both the drift function and the bound function are affine, we are able to compute explicitly I_F and provide explicit expressions for all the quantities involved. Although it should be noted that the logistic diffusion model, as considered in [3], does not fall exactly under the umbrella of our assumptions (the volatility can be zero in that model), we analyze numerically a very similar model.

The rest of the paper is organized as follows. In section 2, we present the model and the control problem, and then we state our main result (Theorem 2.5) which is the solution to the control problem. Section 3 is devoted to the analytical study of the above-mentioned functional (I_F) and it culminates with the proof of its concavity (see Proposition 3.7). In section 4, we obtain an analytical expression for the performance of an arbitrary barrier strategy and use the concavity of I_F to identify the optimal barrier level, while in section 5 we prove that this candidate optimal barrier strategy

is an optimal strategy for the control problem under study. Finally, in section 6, we present a few examples of models and bound functions.

2. Model, control problem, and main result. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a complete and filtered probability space satisfying the *usual conditions*. We assume that the underlying (uncontrolled) process evolves according to the following stochastic differential equation (SDE):

$$dY_t = \mu(Y_t)dt + \sigma(Y_t)dW_t,$$

where $(W_t)_{t \geq 0}$ denotes a standard Brownian motion adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$. Assumptions on the functions μ and σ will be provided later; for now, we assume they are regular enough for the existence of a unique strong solution to the above SDE.

As we are interested in absolutely continuous control process/strategies, we identify each strategy with its adapted density process $\ell = (\ell_t)_{t \geq 0}$, which is the rate of withdrawals. Then, the associated controlled process is given by

$$(2.1) \quad dX_t^\ell = \mu(X_t^\ell)dt + \sigma(X_t^\ell)dW_t - \ell_t dt,$$

where $t \mapsto \int_0^t \ell_s ds$ represents the amount of withdrawals made up to time t . Note that, for any strategy ℓ , the process X^ℓ has continuous sample paths, almost surely.

Let $F: [0, \infty) \rightarrow \mathbb{R}$ be a nondecreasing and concave function such that $F(0) \geq 0$; it is the bound function. As in [13], we say that a control strategy, as defined above, is *admissible* if its density process is such that $\ell_t \in [0, F(X_t^\ell)]$ for all $t \geq 0$. Accordingly, we denote the set of admissible strategies/density rates by Π .

In what follows, we will write $\mathbb{E}_x$ as the associated expectation operator when X^ℓ starts from $X_0^\ell = x \in \mathbb{R}$, and $\mathbb{E}$ when it starts from $X_0^\ell = 0$. We will also use the notation $\mathbb{E}_x[V; A] = \mathbb{E}_x[V\mathbf{1}_A]$ for a random variable V and an event A .

Finally, given a time-preference rate $q > 0$, we define the performance function of a strategy $\ell \in \Pi$ by

$$J(x, \ell) = \mathbb{E}_x \left[\int_0^{\tau_0^\ell} e^{-qt} \ell_t dt \right], \quad x \in \mathbb{R},$$

where $\tau_0^\ell = \inf\{t \geq 0: X_t^\ell \leq 0\}$ is the termination time. Note that $J(x, \ell) = 0$ for all $x \leq 0$ and for any $\ell \in \Pi$.

The objective of our control problem is to maximize this performance function over all admissible strategies. More precisely, we want to find an expression for the value function

$$V(x) := \sup_{\ell \in \Pi} J(x, \ell), \quad x \in \mathbb{R},$$

and we want to find an optimal control strategy ℓ^* , i.e., an admissible strategy such that $V(x) = J(x, \ell^*)$ for all $x \in \mathbb{R}$.

We will reach the above-mentioned objectives of computing $V(x)$ and finding an optimal strategy, under the following standing assumptions.

Hypothesis 2.1.

1. The drift function $\mu: [0, \infty) \rightarrow \mathbb{R}$ is concave, and it is such that $\mu(0) > 0$. In addition, we assume that $\mu'(0+) < q$.
2. The diffusion function $\sigma: [0, \infty) \rightarrow \mathbb{R}$ is Lipschitz continuous, and it is bounded away from zero, i.e., there exists $\epsilon > 0$ such that $\sigma(x) \geq \epsilon$ for all $x \in [0, \infty)$.
3. The bound function $F: [0, \infty) \rightarrow \mathbb{R}$ is a nondecreasing and concave function such that $F(0) \geq 0$.

First, note that these assumptions are sufficient for the existence of unique strong solutions X^ℓ to the associated SDEs, as given in (2.1); see, e.g., Theorem IV.3.1 in [10].

The assumption on μ has appeared before in the literature. For example, see Remark 2.1 in [15] and Condition 2 in [19]. In our problem, the condition on the derivative of μ guarantees the finiteness of important mathematical objects (see Proposition 3.1), while main steps in our proofs rely on the concavity assumption.

The assumption on F has also appeared before in the literature. For example, in [13], it is used to solve the same control problem but with a simple Brownian model. Note that the concavity of F is a technical assumption that is needed in various proofs of section 3 and to prove the optimality of a barrier strategy. Moreover, the concavity of a related function has also appeared in another instance of de Finetti's dividend problem with absolutely continuous dividends; see [9] for further details.

Refraction strategies, an expectation, and differential operators. The experienced reader is expecting a bang-bang strategy to be optimal for this control problem. In this direction, we now define the family of refraction strategies; in [13], these strategies were called *barrier mean-reverting strategies*.

More precisely, for a given $b \geq 0$, we define the refraction strategy with barrier b by the following density process:

$$(2.2) \quad \ell_t^b = F(X_t^b) \mathbf{1}_{\{X_t^b \geq b\}}, \quad t \geq 0,$$

where $X^b = (X_t^b)_{t \geq 0}$ is given by the solution of

$$(2.3) \quad dX_t^b = \left(\mu(X_t^b) - F(X_t^b) \mathbf{1}_{\{X_t^b \geq b\}} \right) dt + \sigma(X_t^b) dW_t.$$

This is indeed a specific example of the SDE given in (2.1). The term *refraction* comes from the fact that the drift function is either $\mu(\cdot)$ or $\mu(\cdot) - F(\cdot)$, whether the process is below or above b .

We denote the performance function of the refraction strategy at level b by

$$(2.4) \quad J_b(x) = \mathbb{E}_x \left[\int_0^{\tau_0^b} e^{-qt} F(X_t^b) \mathbf{1}_{\{X_t^b \geq b\}} dt \right], \quad x \in \mathbb{R},$$

where $\tau_0^b = \inf\{t \geq 0: X_t^b \leq 0\}$.

Remark 2.2. Note that, for any $b \geq 0$, the process X^b is the unique strong solution of (2.3) since its drift function is a piecewise Lipschitz continuous function; see, e.g., [12]. In particular, a refraction strategy at level b is admissible.

If $b = 0$, then X^0 is the controlled process associated with the strategy consisting in withdrawing at the maximal rate until the termination time. Note that, up to its termination time τ_0^0 , this process has the same distribution as $X = (X_t)_{t \geq 0}$ given by

$$(2.5) \quad dX_t = (\mu(X_t) - F(X_t))dt + \sigma(X_t)dW_t.$$

Now, let us define one of the key functions in our study:

$$(2.6) \quad I_F(x) = \mathbb{E}_x \left[\int_0^\infty e^{-qt} F(X_t) dt \right], \quad x \in \mathbb{R}.$$

In order for I_F to be well defined, we must specify the behavior of X in $(-\infty, 0)$.

Remark 2.3 (Extension of μ, σ, F on $\mathbb{R}$). For $x \in (-\infty, 0)$, set

$$\mu(x) = \mu(0) + \mu'(0+)x, \quad F(x) = F(0) + F'(0+)x, \quad \sigma(x) = \sigma(0),$$

where $\mu'(0+)$ and $F'(0+)$ are right derivatives at 0. Note that μ and F are now concave functions on $\mathbb{R}$, while σ is Lipschitz continuous on $\mathbb{R}$.

The function I_F is of paramount importance in our approach. As alluded to above, the main challenge in our approach is to obtain some analytical properties for I_F . This task is undertaken in section 3.

Finally, denote the infinitesimal generator associated with X by

$$\mathcal{L}_F = \frac{\sigma^2(x)}{2} \frac{d^2}{dx^2} + (\mu(x) - F(x)) \frac{d}{dx}.$$

Similarly, we write $\mathcal{L}$ for the infinitesimal generator associated with Y or, equivalently, corresponding to X with $F \equiv 0$. As is well known (see, e.g., [6]), we can denote by ψ_q the positive and increasing fundamental solution of

$$(\mathcal{L} - q)u(x) = 0, \quad x \in (0, \infty),$$

with initial conditions $u(0) = 0$ and $u'(0+) = 1$. Similarly, let $\varphi_{q,F}$ be the positive and decreasing fundamental solution of

$$(\mathcal{L}_F - q)u(x) = 0, \quad x \in (0, \infty),$$

with boundary conditions $u(0) = 1$ and $\lim_{x \rightarrow \infty} u(x) = 0$. It is also well known that these functions appear in first-passage identities related to the corresponding process. In particular, for $b \geq 0$ and $0 \leq x \leq b$, we have

$$(2.7) \quad \mathbb{E}_x [e^{-q\kappa_b}; \kappa_b < \kappa_0] = \frac{\psi_q(x)}{\psi_q(b)},$$

where $\kappa_b = \inf\{t \geq 0: Y_t \geq b\}$ and $\kappa_0 = \inf\{t \geq 0: Y_t \leq 0\}$, while for $x \geq b$, we have

$$(2.8) \quad \mathbb{E}_x [e^{-q\tau_b}] = \frac{\varphi_{q,F}(x)}{\varphi_{q,F}(b)},$$

where $\tau_b = \inf\{t \geq 0: X_t \leq b\}$.

Here are useful properties that we will use repeatedly in the rest of the paper.

LEMMA 2.4 (Lemma 2.4 in [7]). *Under Hypothesis 2.1, ψ_q is strictly concave-convex on $(0, \infty)$ with a unique inflection point $\hat{b} > 0$ and $\varphi_{q,F}$ is strictly convex on $(0, \infty)$.*

We now state our main result.

THEOREM 2.5. *Under Hypothesis 2.1, there exists a unique optimal refraction level $b^* \geq 0$ such that the refraction strategy ℓ^{b^*} is optimal for the control problem. Consequently,*

$$V(x) = \mathbb{E}_x \left[\int_0^{\tau_0^*} e^{-qt} F(X_t^{b^*}) \mathbf{1}_{\{X_t^{b^*} \geq b^*\}} dt \right], \quad x \geq 0,$$

where X^{b^*} satisfies the SDE (2.3) and $\tau_0^* = \inf\{t \geq 0: X_t^{b^*} \leq 0\}$. Also, V is a twice continuously differentiable function characterized by the following dichotomy:

- If $I'_F(0+) - I_F(0)\varphi'_{q,F}(0+) > 1$, then $b^* > 0$ is the root of

$$\frac{\psi_q(b)}{\psi'_q(b)} = I_F(b) + (1 - I'_F(b)) \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)},$$

and V is a concave function given by

$$V(x) = \begin{cases} \frac{\psi_q(x)}{\psi'_q(b^*)} & \text{for } x \leq b^*, \\ I_F(x) - \varphi_{q,F}(x) \left(\frac{1 - I'_F(b^*)}{\varphi'_{q,F}(b^*)} \right) & \text{for } x \geq b^*. \end{cases}$$

- If $I'_F(0+) - I_F(0)\varphi'_{q,F}(0+) \leq 1$, then $b^* = 0$ and

$$V(x) = I_F(x) - I_F(0)\varphi_{q,F}(x), \quad x \geq 0.$$

The remainder of the paper is devoted to proving Theorem 2.5. In particular, in section 4, we will compute the performance function of an arbitrary refraction strategy.

3. Analytical properties of I_F . As mentioned above, our proof of Theorem 2.5 follows a probabilistic guess-and-verify approach based on the computation of the performance function of an arbitrary refraction strategy. The function I_F appears in its expression and the analytical properties of I_F , especially concavity, are needed to prove the existence of an optimal threshold.

Before moving forward, note that I_F takes finite values.

PROPOSITION 3.1. *For $x \in \mathbb{R}$, we have $|I_F(x)| < \infty$.*

A technical proof of this last proposition is provided in Appendix A.

PROPOSITION 3.2. *Under Hypothesis 2.1, we have the following:*

1. *If μ and F are affine functions, then I_F is affine and given by*

$$(3.1) \quad I_F(x) = \frac{F(0)}{q} + \frac{F'(0)}{q(q - \mu'(0) + F'(0))}(qx + \mu(0) - F(0)), \quad x \in \mathbb{R}.$$

2. *In general, for $x \leq 0$, we have*

$$(3.2) \quad I_F(x) = \alpha_0 + \beta_0 x + (I_F(0) - \alpha_0) \phi_{q,F}(x),$$

where $\beta_0 := \frac{F'(0+)}{q - \mu'(0+) + F'(0+)}$, $\alpha_0 := \frac{\mu(0)}{q} \beta_0 + \frac{F(0)}{q} (1 - \beta_0)$, and $\phi_{q,F}$ is the unique increasing solution of

$$(\mathcal{L}_F - q)u(x) = 0, \quad x \in (-\infty, 0),$$

with boundary conditions $u(0) = 1$ and $\lim_{x \rightarrow -\infty} u(x) = 0$.

Proof. The expression for I_F in the affine case is straightforward and left to the reader. In the general case, i.e., under Hypothesis 2.1 together with the extensions on $(-\infty, 0)$ in Remark 2.3, let us consider the Ornstein–Uhlenbeck process $\tilde{X}$ given by

$$d\tilde{X}_t = [\mu(0) - F(0) + (\mu'(0+) - F'(0+)) \tilde{X}_t] dt + \sigma(0) dW_t.$$

Note that if $x \leq 0$, then $\tilde{X}$ and X are equal in law (with respect to $\mathbb{P}_x$) up to time $\rho_0 := \inf\{t \geq 0 : \tilde{X}_t \geq 0\}$. In addition, as for (2.8), if $x \leq 0$, then we have $\mathbb{E}_x[e^{-q\rho_0}] = \phi_{q,F}(x)$, where $\phi_{q,F}$ is the unique increasing solution of

$$(\mathcal{L}_F - q)u(x) = 0, \quad x \in (-\infty, 0),$$

with boundary conditions $u(0) = 1$ and $\lim_{x \rightarrow -\infty} u(x) = 0$.

Successive applications of the strong Markov property give, for $x \leq 0$,

$$I_F(x) = \tilde{I}_F(x) - \tilde{I}_F(0)\phi_{q,F}(x) + I_F(0)\phi_{q,F}(x),$$

where $\tilde{I}_F(x) := \mathbb{E}_x[\int_0^\infty e^{-qt}(F(0) + F'(0+)\tilde{X}_t)dt]$. From the affine case, we have $\tilde{I}_F(x) = \alpha_0 + \beta_0 x$, with $\alpha_0 = \frac{\mu(0)}{q}\beta_0 + \frac{F(0)}{q}(1 - \beta_0)$, and the result follows. $\square$

PROPOSITION 3.3. *The function I_F is twice continuously differentiable on $\mathbb{R}$. Moreover, I_F satisfies $(\mathcal{L}_F - q)I_F(x) = -F(x)$ for $x \in \mathbb{R}$.*

Proof. For $x \geq 0$, we have

$$\mathbb{E}_x[e^{-q\tau_0}] = \frac{\varphi_{q,F}(x)}{\varphi_{q,F}(0)} = \varphi_{q,F}(x),$$

where $\tau_0 = \inf\{t \geq 0: X_t \leq 0\}$. This is a special case of (2.8). Consequently, an application of the strong Markov property yields, for $x \geq 0$,

$$I_F(x) = J_0(x) + I_F(0)\varphi_{q,F}(x),$$

where $J_0(x) = \mathbb{E}_x[\int_0^{\tau_0} e^{-qt}F(X_t)dt]$ is the performance function of the refraction strategy with threshold level $b = 0$.

On one hand, we have that $\varphi_{q,F} \in \mathcal{C}^2(0, \infty)$. On the other hand, since the functional J_0 is obtained by an application of the resolvent operator (killed at zero) of X to the bound function F (see, e.g., equations (18)–(20) in [2]), we have that $J_0 \in \mathcal{C}^2(0, \infty)$ is a classical solution to the following second-order initial value problem:

$$\begin{cases} (\mathcal{L}_F - q)J_0(x) = -F(x), & x \in (0, \infty), \\ J_0(0) = 0. \end{cases}$$

As a consequence, we have also that $I_F \in \mathcal{C}^2(0, \infty)$ and

$$(\mathcal{L}_F - q)I_F(x) = -F(x), \quad x \in (0, \infty).$$

From the expression of I_F on $(-\infty, 0)$ given in (3.2) of Proposition 3.2, we easily deduce that

$$(\mathcal{L}_F - q)I_F(x) = -F(x), \quad x \in (-\infty, 0).$$

So far, we have shown that $I_F \in \mathcal{C}^2(\mathbb{R} \setminus \{0\})$. So, to conclude, it suffices to show that $I'_F(0-) = I'_F(0+)$ and $I''_F(0-) = I''_F(0+)$. Let us proceed by contradiction.

First, suppose $I'_F(0+) < I'_F(0-)$, and let $d \in (I'_F(0+), I'_F(0-))$. For an arbitrary $M > 0$, define the function $\pi_M(x) := I_F(0) + dx - \frac{1}{2}Mx^2$ for $x \in \mathbb{R}$. Note that $\pi_M(0) = I_F(0)$ and, since $d \in (I'_F(0+), I'_F(0-))$, by continuity, there exists $\epsilon_M > 0$ such that $\pi_M(x) \geq I_F(x)$ for all $x \in [-\epsilon_M, \epsilon_M]$.

Let $h > 0$, and set $\tau_M = \inf\{t \geq 0: X_t \notin [-\epsilon_M, \epsilon_M]\} \wedge h$. Using the strong Markov property, we have

$$\begin{aligned} \frac{1}{h}\pi_M(0) &= \frac{1}{h}I_F(0) = \frac{1}{h}\mathbb{E}\left[\int_0^{\tau_M} e^{-qt}F(X_t)dt + e^{-q\tau_M}I_F(X_{\tau_M})\right] \\ &\leq \frac{1}{h}\mathbb{E}\left[\int_0^{\tau_M} e^{-qt}F(X_t)dt + e^{-q\tau_M}\pi_M(X_{\tau_M})\right]. \end{aligned}$$

Using Itô's formula to expand the term $e^{-q\tau_M}\pi_M(X_{\tau_M})$ we obtain

$$0 \leq \frac{1}{h} \mathbb{E} \left[\int_0^{\tau_M} e^{-qt} (F(X_t) + (\mu(X_t) - F(X_t))(d - MX_t) - \frac{1}{2}\sigma^2(X_t)M - q\pi_M(X_t)) dt \right].$$

By taking the limit as $h \downarrow 0$, we obtain $q\pi_M(0) - (\mu(0) - F(0))d + \frac{1}{2}\sigma^2(0)M - F(0) \leq 0$. Taking M large enough leads to a contradiction. So, we must have $I'_F(0+) \geq I'_F(0-)$. In a very similar way, we can prove the reverse inequality. Hence $I_F \in \mathcal{C}^1(\mathbb{R})$.

Recall that $(\mathcal{L}_F - q)I_F = -F$ on $\mathbb{R} \setminus \{0\}$. Taking the limits when $x \downarrow 0$ and $x \uparrow 0$, and using the continuity at zero of the functions, we obtain

$$\begin{aligned} (\mu(0) - F(0))I'_F(0-) + \frac{1}{2}\sigma^2(0)I''_F(0-) - qI_F(0-) + F(0) &= 0, \\ (\mu(0) - F(0))I'_F(0+) + \frac{1}{2}\sigma^2(0)I''_F(0+) - qI_F(0+) + F(0) &= 0. \end{aligned}$$

We have shown above that $I_F(0+) = I_F(0)$ and $I'_F(0-) = I'_F(0+)$ and, from (3.2) of Proposition 3.2, we easily deduce that $I_F(0-) = I_F(0)$. Therefore, from the above two equalities, we now have that $I''_F(0-) = I''_F(0+)$. $\square$

PROPOSITION 3.4. *For $x \in \mathbb{R}$, we have $I'_F(x) \in [0, 1]$.*

Proof. The monotonicity of I_F follows directly from the fact F is increasing and the comparison theorem for diffusions (see, e.g., [10, Chap. VI, Theorem 1.1]).

Fix $x \in \mathbb{R}$ and $T > 0$. By an application of Dynkin's formula,

$$\mathbb{E} [e^{-qT} X_T^x] = x + \mathbb{E} \left[\int_0^T e^{-qt} (\mu(X_t^x) - F(X_t^x) - qX_t^x) dt \right],$$

in which the notation X^x is used (only) in this proof to emphasize that $X_0^x = x$. Taking the limit as $T \rightarrow \infty$, using Lemma A.1, and rearranging terms, we get

$$(3.3) \quad I_F(x) = x + \mathbb{E} \left[\int_0^\infty e^{-qt} (\mu(X_t^x) - qX_t^x) dt \right], \quad x \in \mathbb{R}.$$

For $h > 0$, since I_F is nondecreasing, we have $I_F(x+h) - I_F(x) \geq 0$. Moreover,

$$I_F(x+h) - I_F(x) = h + \mathbb{E} \left[\int_0^\infty e^{-qt} (\mu(X_t^{x+h}) - \mu(X_t^x) - q(X_t^{x+h} - X_t^x)) dt \right] \leq h,$$

where we used that, for $y_1 \leq y_2$, we have $\mu(y_2) - \mu(y_1) \leq q(y_2 - y_1)$ and $X_t^{x+h} \geq X_t^x$ for all $t \geq 0$. Thus, we have obtained $\frac{I_F(x+h) - I_F(x)}{h} \leq 1$, and as I'_F exists, the result follows by taking the limit as $h \rightarrow 0$. $\square$

We now prove that the growth of I_F is at most linear, with an explicit estimation which will be key to proving the concavity.

LEMMA 3.5. *Let $\beta_\xi = \frac{F'(\xi)}{q - \mu'(\xi) + F'(\xi)}$ and $\alpha_\xi = \frac{\beta_\xi}{q} [\mu(\xi) - q\xi + (q - \mu'(\xi)) \frac{F(\xi)}{F'(\xi)}]$ for $\xi \in \mathbb{R}$. We have*

$$(3.4) \quad I_F(x) \leq \alpha_\xi + \beta_\xi x, \quad x \in \mathbb{R}.$$

In addition, for $x \leq 0$, we have

$$(3.5) \quad I_F(0) + \beta_0 x \leq I_F(x) \leq \alpha_0 + \beta_0 x.$$

Proof. First, note that, for $\xi < 0$, we have $\beta_\xi = \beta_0$ and $\alpha_\xi = \alpha_0$.

Let $\xi \in \mathbb{R}$, $x \in \mathbb{R}$. For simplicity, we write β for β_ξ . Using (3.3), we have

$$\beta x = I_F(x) + \mathbb{E} \left[\int_0^\infty e^{-qt} \left(\beta \left[qX_t^x - \mu(X_t^x) + \left(\frac{\beta-1}{\beta} \right) F(X_t^x) \right] \right) dt \right].$$

First, note that $\beta > 0$ and $\frac{\beta-1}{\beta} = \frac{\mu'(\xi)-q}{F'(\xi)} \leq 0$. Second, since F is concave on $\mathbb{R}$, we have $F(X_t^x) \leq F(\xi) + F'(\xi)(X_t^x - \xi)$. Consequently,

$$\begin{aligned} (3.6) \quad \beta x - I_F(x) &\geq \mathbb{E} \left[\int_0^\infty e^{-qt} \left(\beta \left[qX_t^x - \mu(X_t^x) - (q - \mu'(\xi)) \left(\frac{F(\xi)}{F'(\xi)} + X_t^x - \xi \right) \right] \right) dt \right] \\ &= \mathbb{E} \left[\int_0^\infty e^{-qt} \left(\beta \left[-\mu(X_t^x) - (q - \mu'(\xi)) \frac{F(\xi)}{F'(\xi)} + q\xi + \mu'(\xi)(X_t^x - \xi) \right] \right) dt \right] \\ &\geq \int_0^\infty e^{-qt} \left(\beta \left[q\xi - \mu(\xi) - (q - \mu'(\xi)) \frac{F(\xi)}{F'(\xi)} \right] \right) dt, \end{aligned}$$

where, in the last inequality, we used the fact that $\mu(X_t^x) \leq \mu(\xi) + \mu'(\xi)(X_t^x - \xi)$, since μ is a concave function on $\mathbb{R}$. It follows from (3.6) that $\beta x - I_F(x) \geq -\alpha_\xi$.

Finally, for the proof of (3.5), recall from (3.2) that, for $x \leq 0$,

$$I_F(x) = \alpha_0 + \beta_0 x + (I_F(0) - \alpha_0) \phi_{q,F}(x) = I_F(0) + \beta_0 x + (\alpha_0 - I_F(0)) (1 - \phi_{q,F}(x)).$$

In addition, it follows from (3.4) by taking $\xi = 0$ that $I_F(0) \leq \alpha_0$. We deduce easily that, for $x \leq 0$, $I_F(0) + \beta_0 x \leq I_F(x) \leq \alpha_0 + \beta_0 x$. $\square$

Remark 3.6. Under Hypothesis 2.1, we have that $\xi \mapsto \beta_\xi$ is nonincreasing.

Let us now investigate the concavity of I_F . First, note that concavity on $(-\infty, 0)$ easily follows from (3.2). Indeed, for $x \leq 0$,

$$I_F(x) = I_F(0) + \beta_0 x + (\alpha_0 - I_F(0)) (1 - \phi_{q,F}(x)).$$

As $\phi_{q,F}$ is convex (see Lemma 2.4) and $\alpha_0 \geq I_F(0)$, it follows easily that I_F is concave on $(-\infty, 0)$. Furthermore, as $I_F \in \mathcal{C}^2(\mathbb{R})$, we can deduce that $I_F''(0) \leq 0$.

PROPOSITION 3.7. *The function I_F is concave on $\mathbb{R}$.*

Proof. By Proposition 3.2, if μ and F are affine, I_F is affine and therefore concave, so assume that μ or F is not affine. Let U be the concave envelope of I_F given by

$$U(z) = \inf \{ \Phi(z) : \Phi \text{ is concave and } \Phi \geq I_F \}, \quad z \in \mathbb{R},$$

which is itself concave. See Appendix B for more details on concave envelopes.

To obtain the concavity of I_F , we prove that $I_F = U$. Therefore, it suffices to prove that $I_F \geq U$. Let us proceed by contradiction and suppose there exists $x_0 > 0$ such that $U(x_0) > I_F(x_0)$. As $I_F \in \mathcal{C}^2(\mathbb{R})$, there exist $-\infty \leq \hat{x} < \hat{y} \leq \infty$ such that $x_0 \in (\hat{x}, \hat{y})$, $U(z) > I_F(z)$ for all $z \in (\hat{x}, \hat{y})$, and $U(\hat{x}) = I_F(x_0)$ if $\hat{x} > -\infty$ (resp., $U(\hat{y}) = I_F(x_0)$ if $\hat{y} < \infty$). We will study separately the following cases:

- (1) $-\infty < \hat{x} < \hat{y} < +\infty$;
- (2) $\hat{x} > -\infty$ and $\hat{y} = +\infty$;
- (3) $\hat{x} = -\infty$ and $\hat{y} < +\infty$;
- (4) $\hat{x} = -\infty$ and $\hat{y} = +\infty$.

First, we will prove, in cases (1)–(3), that $(q - \mathcal{L}_F)U(z) \leq F(z)$ for $z \in (\hat{x}, \hat{y})$. For case (1), from Lemmas B.1 and B.2, we have that U is affine on $(\hat{x}, \hat{y})$ and

$$U(\hat{x}) = I_F(\hat{x}), U(\hat{y}) = I_F(\hat{y}), U' = I'_F(\hat{x}) = I'_F(\hat{y}), I''_F(\hat{x}) \leq 0, I''_F(\hat{y}) \leq 0.$$

Thus, using also that $(q - \mathcal{L}_F)I_F = F$ on $\mathbb{R}$ from Proposition 3.3, we obtain

$$\begin{aligned} (3.7) \quad (q - \mathcal{L}_F)U(\hat{x}) - F(\hat{x}) &= qI_F(\hat{x}) - (\mu(\hat{x}) - F(\hat{x}))I'_F(\hat{x}) - F(\hat{x}) \\ &= (q - \mathcal{L}_F)I_F(\hat{x}) - F(\hat{x}) + \frac{\sigma^2(\hat{x})}{2}I''_F(\hat{x}) = \frac{\sigma^2(\hat{x})}{2}I''_F(\hat{x}) \leq 0. \end{aligned}$$

Similarly, we have $(q - \mathcal{L}_F)U(\hat{y}) - F(\hat{y}) \leq 0$.

For $z \in (\hat{x}, \hat{y})$, we have $z = \lambda\hat{x} + (1 - \lambda)\hat{y}$ for some $\lambda \in (0, 1)$. Using the concavity of both μ and F and using Proposition 3.4, we have

$$\begin{aligned} (q - \mathcal{L}_F)U(z) - F(z) &= q(\lambda U(\hat{x}) + (1 - \lambda)U(\hat{y})) - I'_F(\hat{x})\mu(z) + F(z)(I'_F(\hat{x}) - 1) \\ &\leq q(\lambda U(\hat{x}) + (1 - \lambda)U(\hat{y})) - I'_F(\hat{x})(\lambda\mu(\hat{x}) + (1 - \lambda)\mu(\hat{y})) \\ &\quad + (I'_F(\hat{x}) - 1)(\lambda F(\hat{x}) + (1 - \lambda)F(\hat{y})) \\ &\leq \lambda((q - \mathcal{L}_F)U(\hat{x}) - F(\hat{x})) + (1 - \lambda)((q - \mathcal{L}_F)U(\hat{y}) - F(\hat{y})) \\ &\leq 0, \end{aligned}$$

where the last inequality is a direct consequence of inequality (3.7) and its equivalent with $\hat{x}$ replaced by $\hat{y}$. It follows that $(q - \mathcal{L}_F)U(z) \leq F(z)$ for $z \in [\hat{x}, \hat{y}]$.

For case (2), from Lemmas B.1 and B.2, we know that U is affine on $(\hat{x}, \infty)$ and

$$(3.8) \quad U(\hat{x}) = I_F(\hat{x}), U'(\hat{x}) = I'_F(\hat{x}), I''_F(\hat{x}) \leq 0.$$

In particular, we have $U(z) = I_F(\hat{x}) + I'_F(\hat{x})(z - \hat{x})$ for $z \in (\hat{x}, \infty)$. Then,

$$\begin{aligned} (q - \mathcal{L}_F)U(z) - F(z) &= qI_F(\hat{x}) + I'_F(\hat{x})(q(z - \hat{x}) - \mu(z) + F(z)) - F(z) \\ &= I'_F(\hat{x})(\mu(\hat{x}) - F(\hat{x})) + \frac{\sigma^2(\hat{x})}{2}I''_F(\hat{x}) + F(\hat{x}) \\ &\quad + I'_F(\hat{x})(q(z - \hat{x}) - \mu(z) + F(z)) - F(z) \end{aligned}$$

for $z \in (\hat{x}, \infty)$, where in the second equality we used that $(q - \mathcal{L}_F)I_F(\hat{x}) = F(\hat{x})$ in order to substitute for $qI_F(\hat{x})$. After rearranging terms, we get

$$\begin{aligned} (3.9) \quad (q - \mathcal{L}_F)U(z) - F(z) &= \frac{\sigma^2(\hat{x})}{2}I''_F(\hat{x}) \\ &\quad + I'_F(\hat{x}) \left(q(z - \hat{x}) - (\mu(z) - \mu(\hat{x})) + \frac{I'_F(\hat{x}) - 1}{I'_F(\hat{x})}(F(z) - F(\hat{x})) \right). \end{aligned}$$

Now, let us obtain a suitable bound for the term $\frac{I'_F(\hat{x}) - 1}{I'_F(\hat{x})}$. To this end, we notice by Lemma 3.5 and by the definition of the concave envelope that, for all $\xi \in \mathbb{R}$,

$$U(x) \leq \alpha_\xi + \beta_\xi x, \quad x \in \mathbb{R}.$$

Since U is affine with slope $I'_F(\hat{x})$ on $(\hat{x}, \infty)$, we deduce $I'_F(\hat{x}) \leq \beta_\xi$ for any $\xi \in \mathbb{R}$.

Using the definition of β_ξ and $-q + \mu'(\xi) < 0$, we have that

$$(3.10) \quad \frac{I'_F(\hat{x}) - 1}{I'_F(\hat{x})} \leq \frac{-q + \mu'(\xi)}{F'(\xi)}.$$

Applying the inequality (3.10) with $\xi = z$ in (3.9) yields

$$\begin{aligned} (q - \mathcal{L}_F)U(z) - F(z) &\leq \frac{\sigma^2(\hat{x})}{2} I''_F(\hat{x}) + I'_F(\hat{x})(z - \hat{x}) \left(q - \frac{\mu(z) - \mu(\hat{x})}{z - \hat{x}} - \left(\frac{q - \mu'(z)}{F'(z)} \right) \frac{F(z) - F(\hat{x})}{z - \hat{x}} \right) \\ &\leq \frac{\sigma^2(\hat{x})}{2} I''_F(\hat{x}) + I'_F(\hat{x})(z - \hat{x}) \left(q - \mu'(z) - \left(\frac{q - \mu'(z)}{F'(z)} \right) F'(z) \right) = \frac{\sigma^2(\hat{x})}{2} I''_F(\hat{x}) \leq 0, \end{aligned}$$

where we used that I_F and F are increasing, that μ and F are concave, and (3.8). This allows us to conclude that $(q - \mathcal{L}_F)U(z) \leq F(z)$ for $z \in (\hat{x}, +\infty)$.

In case (3), we similarly have $U(z) = I_F(\hat{y}) + I'_F(\hat{y})(z - \hat{y})$ for $z \in (-\infty, \hat{y})$, and $I''_F(\hat{y}) \leq 0$. Using the bounds (3.5) of I_F , we deduce that $U'(\hat{y}) = I'_F(\hat{y}) = \beta_0$. Furthermore, by monotonicity of $\xi \mapsto \beta_\xi$ (see Remark 3.6), for any $\xi \in \mathbb{R}$, $\frac{I'_F(\hat{y}) - 1}{I'_F(\hat{y})} = \frac{-q + \mu'(0+)}{F'(0+)} \geq \frac{-q + \mu'(\xi)}{F'(\xi)}$. As in case (2), we deduce $(q - \mathcal{L}_F)U(z) \leq F(z)$ for $z < \hat{y}$.

At last in case (4), by Lemma B.1 we know that U is affine on $\mathbb{R}$: $U(z) = U(0) + U'(0)z$. Similarly to case (2), we deduce that $U'(0) \leq \beta_\xi$ for any $\xi \in \mathbb{R}$, and $U'(0) = \beta_0$. From monotonicity of $\xi \mapsto \beta_\xi$, we deduce that case (4) can only happen when β_ξ is constant. Under Hypothesis 2.1 this is only true when F' and μ' are constant, which is impossible by hypothesis.

We can now obtain a contradiction in cases (1)–(3). Define the exit time $\hat{T} = \inf\{t \geq 0 : X_t \notin (\hat{x}, \hat{y})\}$. On $\{\hat{T} < +\infty\}$ we have $X_{\hat{T}} \in \{\hat{x}, \hat{y}\}$ in case (1); $X_{\hat{T}} = \hat{x}$ in case (2); and $X_{\hat{T}} = \hat{y}$ in case (3). As $x_0 \in (\hat{x}, \hat{y})$, by the strong Markov property, we have

$$\begin{aligned} (3.11) \quad I_F(x_0) &= \mathbb{E}_{x_0} \left[\int_0^{\hat{T}} e^{-qt} F(X_t) dt + e^{-q\hat{T}} I_F(X_{\hat{T}}); \hat{T} < +\infty \right] \\ &\quad + \mathbb{E}_{x_0} \left[\int_0^{+\infty} e^{-qt} F(X_t) dt; \hat{T} = +\infty \right] \\ &\geq \mathbb{E}_{x_0} \left[\int_0^{\hat{T}} e^{-qt} (q - \mathcal{L}_F)U(X_t) dt + e^{-q\hat{T}} I_F(X_{\hat{T}}) \mathbf{1}_{\hat{T} < +\infty} \right] \\ &= \mathbb{E}_{x_0} \left[\int_0^{\hat{T}} e^{-qt} (q - \mathcal{L}_F)U(X_t) dt + e^{-q\hat{T}} U(X_{\hat{T}}) \mathbf{1}_{\hat{T} < +\infty} \right], \end{aligned}$$

where we used that, on $\{\hat{T} < \infty\}$, $I_F(\hat{x}) = U(\hat{x})$ and/or $I_F(\hat{y}) = U(\hat{y})$.

In order to conclude, we analyze the term on the right side of (3.11). Let $\tau_n := \hat{T} \wedge n$. Then, using Itô's formula we get

$$e^{-q\tau_n} U(X_{\tau_n}) - U(x_0) = \int_0^{\tau_n} e^{-qt} (\mathcal{L}_F - q)U(X_t) dt + \int_0^{\tau_n} \sigma(X_t) U'(X_t) dW_t.$$

By taking expectations on both sides, we obtain

$$(3.12) \quad \mathbb{E}_{x_0} [e^{-q\tau_n} U(X_{\tau_n})] - U(x_0) = \mathbb{E}_{x_0} \left[\int_0^{\tau_n} e^{-qt} (\mathcal{L}_F - q)U(X_t) dt \right].$$

Observe that we can expand the term $\mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})]$ as

$$\mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})] = \mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})\mathbf{1}_{\{\hat{T} < +\infty\}}] + \mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})\mathbf{1}_{\{\hat{T} = +\infty\}}],$$

where it follows that $\mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})\mathbf{1}_{\{\hat{T} = +\infty\}}] \rightarrow 0$ as $n \rightarrow \infty$ due to the fact that U is affine and to Lemma A.1, and $\mathbb{E}_{x_0}[e^{-q\tau_n}U(X_{\tau_n})\mathbf{1}_{\{\hat{T} < +\infty\}}]$ converges to $\mathbb{E}_{x_0}[e^{-q\hat{T}}U(X_{\hat{T}})\mathbf{1}_{\{\hat{T} < +\infty\}}]$ as $n \rightarrow \infty$. Taking the limit in (3.12) as $n \rightarrow \infty$, we get

$$\mathbb{E}_{x_0}[e^{-q\hat{T}}U(X_{\hat{T}})\mathbf{1}_{\{\hat{T} < +\infty\}}] - U(x_0) = \mathbb{E}_{x_0}\left[\int_0^{\hat{T}} e^{-qt}(\mathcal{L}_F - q)U(X_t)dt\right],$$

and, by rearranging terms, we obtain

$$\mathbb{E}_{x_0}\left[\int_0^{\hat{T}} e^{-qt}(q - \mathcal{L}_F)U(X_t)dt + e^{-q\hat{T}}U(X_{\hat{T}})\mathbf{1}_{\{\hat{T} < +\infty\}}\right] = U(x_0),$$

where it follows easily that $I_F(x_0) \geq U(x_0)$, a contradiction. $\square$

4. A candidate optimal strategy. In this section, we study the family of refraction strategies which are defined by (2.2) and (2.3). Despite the fact that we consider a general diffusion model, the rest of this section follows closely the general methodology already used in [13] for a Brownian motion with drift.

We provide an intuitive justification as to why we are interested in proving the optimality of refraction strategies. To this end, we observe that the Hamilton–Jacobi–Bellman (HJB) equation associated with our control problem is given by

$$(4.1) \quad \begin{cases} (\mathcal{L} - q)u(x) + \sup_{c \in [0, F(x)]} c(1 - u'(x)) = 0, & x \in (0, \infty), \\ u(0) = 0. \end{cases}$$

From the HJB equation, if u is a classical solution of (4.1), then

$$\begin{cases} (\mathcal{L} - q)u(x) = 0 & \text{if } u'(x) \geq 1, \\ (\mathcal{L}_F - q)u(x) = -F(x) & \text{if } u'(x) < 1. \end{cases}$$

These variational inequalities have the following interpretation:

- If $u'(x) \geq 1$, then the supremum in (4.1) is attained for $c = 0$ and no withdrawals are made.
- If $u'(x) < 1$, then the supremum in (4.1) is attained for $c = F(x)$ and withdrawals are made at the maximal possible rate.

Our goal is to find a classical solution of (4.1) and characterize the regions

$$\mathcal{D}_0 := \{x \in [0, \infty) : u'(x) \geq 1\} \quad \text{and} \quad \mathcal{D}_F := \{x \in [0, \infty) : u'(x) < 1\}.$$

In the context of refraction strategies, given a threshold $b \geq 0$, we notice that the region where no withdrawals are made is the interval $[0, b]$; whereas the region where withdrawals are made at the maximal possible rate is the interval (b, ∞) . Hence, we would like to prove the existence of a unique threshold b^* such that its associated performance function is a classical solution of (4.1) and satisfies $\mathcal{D}_0 = [0, b^*]$ and $\mathcal{D}_F = (b^*, \infty)$.

First, we will derive an expression for the performance function J_b (see (2.4)) using the exit identities (2.7) and (2.8). Fix $b > 0$, and define $\kappa_b = \inf\{t \geq 0 : Y_t \geq b\}$

and $\kappa_0 = \inf\{t \geq 0: Y_t \leq 0\}$. Then, for $x \in [0, b]$, using the fact that X^b and Y are equal until they exit the interval $(0, b)$ and the strong Markov property of Y , we have

$$(4.2) \quad J_b(x) = \mathbb{E}_x \left[e^{-q\kappa_b}; \kappa_b < \kappa_0 \right] J_b(b) = \frac{\psi_q(x)}{\psi_q(b)} J_b(b).$$

Similarly, for $x \in (b, \infty)$, using repeatedly the strong Markov property,

$$(4.3) \quad \begin{aligned} J_b(x) &= \mathbb{E}_x \left[\int_0^{\tau_b} e^{-qt} F(X_t) dt \right] + \mathbb{E}_x \left[e^{-q\tau_b} \right] J_b(b) \\ &= \mathbb{E}_x \left[\int_0^\infty e^{-qt} F(X_t) dt \right] - \mathbb{E}_x \left[\int_{\tau_b}^\infty e^{-qt} F(X_t) dt \right] + \mathbb{E}_x \left[e^{-q\tau_b} \right] J_b(b) \\ &= \mathbb{E}_x \left[\int_0^\infty e^{-qt} F(X_t) dt \right] - \mathbb{E}_x \left[e^{-q\tau_b} \right] \mathbb{E}_b \left[\int_0^\infty e^{-qt} F(X_t) dt \right] + \mathbb{E}_x \left[e^{-q\tau_b} \right] J_b(b) \\ &= I_F(x) + \frac{\varphi_{q,F}(x)}{\varphi_{q,F}(b)} (J_b(b) - I_F(b)). \end{aligned}$$

Using the same approximation procedure as in [13], one can prove that

$$J_b(b) = \psi_q(b) \left(\frac{I'_F(b) \varphi_{q,F}(b) - \varphi'_{q,F}(b) I_F(b)}{\psi'_q(b) \varphi_{q,F}(b) - \varphi'_{q,F}(b) \psi_q(b)} \right).$$

Substituting this value of $J_b(b)$ back into the expressions obtained above, we get

$$J_b(x) = \begin{cases} \psi_q(x) \left(\frac{I'_F(b) \varphi_{q,F}(b) - I_F(b) \varphi'_{q,F}(b)}{\psi'_q(b) \varphi_{q,F}(b) - \psi_q(b) \varphi'_{q,F}(b)} \right) & \text{for } 0 \leq x < b, \\ I_F(x) + \varphi_{q,F}(x) \left(\frac{I'_F(b) \psi_q(b) - I_F(b) \psi'_q(b)}{\psi'_q(b) \varphi_{q,F}(b) - \psi_q(b) \varphi'_{q,F}(b)} \right) & \text{for } x \geq b. \end{cases}$$

We readily see that $J_b \in \mathcal{C}^1(0, \infty)$ for $b > 0$, since $J'_b(b+) = J'_b(b-)$.

On the other hand, for $b = 0$, recall from the proof of Proposition 3.3 that J_0 is a classical solution of

$$\begin{cases} (\mathcal{L}_F - q)u(x) = -F(x), & x \in (0, \infty), \\ u(0) = 0. \end{cases}$$

The next step in our approach consists in finding the *best* refraction strategy. In other words, we want to characterize the optimal refraction level. Mathematically speaking, we want to find a value $b^* \geq 0$ such that $J_{b^*} \geq J_b$, for any $b \geq 0$. In what follows, we will distinguish between the cases $b^* = 0$ and $b^* > 0$.

For reasons that will soon be clear, if $I'_F(0) - I_F(0) \varphi'_{q,F}(0+) \leq 1$, then we set $b^* = 0$. If $I'_F(0) - I_F(0) \varphi'_{q,F}(0+) > 1$, then our goal is to find a value $b^* > 0$ such that $\mathcal{D}_0 = [0, b^*]$ and $\mathcal{D}_F = (b^*, \infty)$. From the discussion at the beginning of this section, we expect that b^* will be such that $J'_{b^*}(b^*) = 1$, which after some algebraic manipulations amounts to finding the root of

$$(4.4) \quad \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - \frac{\psi_q(b)}{\psi'_q(b)} = I'_F(b) \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - I_F(b).$$

The proof that a solution to this last equation exists is provided by the next proposition. In this context, concavity of I_F is a key result.

PROPOSITION 4.1. *If $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) > 1$, then there exists a solution to (4.4) in the interval $(0, \hat{b}]$.*

Proof. We follow the proof of Proposition 4.1 in [13]. Set $g(b) := \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - \frac{\psi_q(b)}{\psi'_q(b)}$ and $h(b) := I'_F(b) \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - I_F(b)$. Since $h(0) \leq g(0)$ by assumption, and both g and h are continuous, by the intermediate value theorem, it suffices to prove that $g(\hat{b}) \leq h(\hat{b})$.

First, since $\hat{b}$ is the unique inflection point of ψ_q , we have $\frac{\psi_q(\hat{b})}{\psi'_q(\hat{b})} = \frac{\mu(\hat{b})}{q}$, hence $g(\hat{b}) = \frac{\varphi_{q,F}(\hat{b})}{\varphi'_{q,F}(\hat{b})} - \frac{\mu(\hat{b})}{q}$. In addition, we have $g(\hat{b}) \leq \frac{-F(\hat{b})}{q}$, due to the convexity of $\varphi_{q,F}$. Now, we focus on the computation of $h(\hat{b})$. Recall that I_F satisfies the nonhomogeneous ODE: $(\mathcal{L}_F - q)I_F(b) + F(b) = 0$. In particular, for $\hat{b}$, and due to the concavity of I_F , we have

$$I_F(\hat{b}) = \frac{\sigma^2(\hat{b})}{2q} I''_F(\hat{b}) + \frac{\mu(\hat{b})}{q} I'_F(\hat{b}) + \frac{F(\hat{b})}{q} (1 - I'_F(\hat{b})) \leq \frac{\mu(\hat{b})}{q} I'_F(\hat{b}) + \frac{F(\hat{b})}{q} (1 - I'_F(\hat{b})).$$

It follows that

$$\begin{aligned} h(\hat{b}) &\geq I'_F(\hat{b}) \frac{\varphi_{q,F}(\hat{b})}{\varphi'_{q,F}(\hat{b})} - \left(\frac{\mu(\hat{b})}{q} I'_F(\hat{b}) + \frac{F(\hat{b})}{q} (1 - I'_F(\hat{b})) \right) \\ &= I'_F(\hat{b}) \left(\frac{\varphi_{q,F}(\hat{b})}{\varphi'_{q,F}(\hat{b})} - \frac{\mu(\hat{b})}{q} \right) - \frac{F(\hat{b})}{q} (1 - I'_F(\hat{b})) \\ &= I'_F(\hat{b}) g(\hat{b}) - \frac{F(\hat{b})}{q} (1 - I'_F(\hat{b})) \geq I'_F(\hat{b}) g(\hat{b}) + g(\hat{b}) (1 - I'_F(\hat{b})) \\ &= g(\hat{b}), \end{aligned}$$

since $I'_F(\hat{b}) \in [0, 1]$. The result follows. $\square$

Note that Proposition 4.1 is an existence result. In what follows, if $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) > 1$, then we take any $b^* \in (0, \hat{b}]$ that is a solution to (4.4). If $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) \leq 1$, then we set $b^* = 0$.

Remark 4.2. At the end of the proof of Theorem 2.5, we will prove that there is a unique solution to (4.4).

Remark 4.3. The condition in Proposition 4.1 is equivalent to $J'_0(0+) > 1$.

If $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) > 1$, then, as discussed above, b^* is a solution to (4.4) and we have $J'_{b^*}(b^*) = 1$. Thus, it follows from (4.2) and (4.3) that $\frac{J_{b^*}(b^*)}{\psi_q(b^*)} = \frac{1}{\psi'_{q'}(b^*)}$ and $\frac{J_{b^*}(b^*) - I_F(b^*)}{\varphi_{q,F}(b^*)} = \frac{1 - I'_F(b^*)}{\varphi'_{q,F}(b^*)}$. So, when $b^* > 0$, we can write

$$(4.5) \quad J_{b^*}(x) = \begin{cases} \frac{\psi_q(x)}{\psi'_q(b^*)}, & 0 \leq x \leq b^*, \\ I_F(x) + \varphi_{q,F}(x) \left(\frac{1 - I'_F(b^*)}{\varphi'_{q,F}(b^*)} \right), & x \geq b^*. \end{cases}$$

5. Verification of optimality. In this section, we complete the proof of Theorem 2.5.

PROPOSITION 5.1. *If $b^* = 0$, then $J'_0(x) \in [0, 1]$ for all $x > 0$. If $b^* > 0$, then J_{b^*} is concave. In both cases, we have that $J_{b^*} \in \mathcal{C}^2(0, \infty)$ and it is a classical solution to the HJB equation given in (4.1).*

Proof. First, note that if $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) \leq 1$, then $I_F(0) \geq \frac{I'_F(0)-1}{\varphi'_{q,F}(0+)}$. Consequently, together with the fact that $J_0(x) = I_F(x) - I_F(0)\varphi_{q,F}(x)$ for $x \in [0, \infty)$, that I_F is concave, and that $\varphi_{q,F}$ is strictly decreasing, we can write, for $x > 0$,

$$J'_0(x) = I'_F(x) - I_F(0)\varphi'_{q,F}(x) \leq I'_F(0) - \frac{I'_F(0)-1}{\varphi'_{q,F}(0+)}\varphi'_{q,F}(0+) = 1.$$

As J_0 is nondecreasing, we conclude that $J'_0(x) \in [0, 1]$ for all $x \in (0, \infty)$.

We have already mentioned that $(\mathcal{L}_F - q)J_0(x) = -F(x)$ for $x \in (0, \infty)$. Thus,

$$\begin{aligned} (\mathcal{L} - q)J_0(x) + \sup_{c \in [0, F(x)]} c(1 - J'_0(x)) &= (\mathcal{L}_F - q)J_0(x) + F(x)J'_0(x) \\ &\quad + F(x)(1 - J'_0(x)) = 0 \end{aligned}$$

for all $x > 0$. In other words, J_0 is a classical solution to the HJB equation in (4.1).

Now, we consider the case $b^* > 0$, i.e., we assume $I'_F(0) - I_F(0)\varphi'_{q,F}(0+) > 1$. Using the representation of J_{b^*} given in (4.5), along with the fact that $b^* \leq \hat{b}$, that $\varphi_{q,F}$ is decreasing, and that $J_{b^*} \in \mathcal{C}^1(0, \infty)$, it follows that J_{b^*} is concave on $\mathbb{R}^+$. It remains to verify that $J_{b^*} \in \mathcal{C}^2(0, \infty)$. To this end, we prove that $J''_{b^*}(b^*-) = J''_{b^*}(b^*+)$.

On one hand, we have $J''_{b^*}(b^*-) = \frac{\psi''_q(b^*)}{\psi'_q(b^*)}$. Note that, by definition, ψ_q is such that $(\mathcal{L} - q)\psi_q(x) = 0$ for $x \in (0, \infty)$, which is equivalent to

$$\frac{\psi''_q(b^*)}{\psi'_q(b^*)} = \frac{2}{\sigma^2(b^*)} \left(q \frac{\psi_q(b^*)}{\psi'_q(b^*)} - \mu(b^*) \right).$$

On the other hand, we have $J''_{b^*}(b^*+) = I''_F(b^*) + \varphi''_{q,F}(b^*)\left(\frac{1-I'_F(b^*)}{\varphi'_{q,F}(b^*)}\right)$. Again, by definition, $\varphi_{q,F}$ is such that $(\mathcal{L}_F - q)\varphi_{q,F}(x) = 0$ for $x > 0$, which is equivalent to

$$(5.1) \quad \frac{\varphi''_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} = \frac{2}{\sigma^2(b^*)} \left(q \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} - (\mu(b^*) - F(b^*)) \right).$$

Similarly, for I_F , we can write

$$(5.2) \quad I''_F(b^*) + \frac{2}{\sigma^2(b^*)}(\mu(b^*) - F(b^*))I'_F(b^*) = \frac{2}{\sigma^2(b^*)}(qI_F(b^*) - F(b^*)).$$

Hence, by substituting (5.1) and (5.2) in the expression for $J''_{b^*}(b^*+)$, we can write

$$\begin{aligned} J''_{b^*}(b^*+) &= I''_F(b^*) + \frac{2}{\sigma^2(b^*)} \left(q \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} - (\mu(b^*) - F(b^*)) \right) (1 - I'_F(b^*)) \\ &= \frac{2q}{\sigma^2(b^*)} \left(I_F(b^*) - \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} I'_F(b^*) \right) + \frac{2q}{\sigma^2(b^*)} \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} - \frac{2}{\sigma^2(b^*)} \mu(b^*) \\ &= \frac{2q}{\sigma^2(b^*)} \left(\frac{\psi_q(b^*)}{\psi'_q(b^*)} - \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} \right) + \frac{2q}{\sigma^2(b^*)} \frac{\varphi_{q,F}(b^*)}{\varphi'_{q,F}(b^*)} - \frac{2}{\sigma^2(b^*)} \mu(b^*) \\ &= \frac{2}{\sigma^2(b^*)} \left(q \frac{\psi_q(b^*)}{\psi'_q(b^*)} - \mu(b^*) \right) = J''_{b^*}(b^*-), \end{aligned}$$

where in the penultimate equality we used Proposition 4.1.

Now, from the smooth-fit condition (4.4) and the concavity of J_{b^*} , we have

$$(5.3) \quad \begin{cases} J'_{b^*}(x) > 1 & \text{if } x \in (0, b^*), \\ J'_{b^*}(x) \leq 1 & \text{if } x \in (b^*, \infty). \end{cases}$$

As a consequence, since we also have

$$(5.4) \quad \begin{cases} (\mathcal{L} - q)J_{b^*}(x) = 0 & \text{if } x \in (0, b^*), \\ (\mathcal{L} - q)J_{b^*}(x) = -F(x)(1 - J'_{b^*}(x)) & \text{if } x \in (b^*, \infty), \end{cases}$$

it follows that J_{b^*} is a classical solution of the HJB equation given in (4.1). $\square$

5.1. Proof of Theorem 2.5. It was proved in Proposition 5.1 that J_{b^*} is a classical solution of the associated HJB equation. All that is left to prove is that $J_{b^*}(x) \geq V(x)$ for all $x \geq 0$.

First note that from (5.3) and (5.4), for $x > 0$ and $u \in [0, F(x)]$, we have

$$(5.5) \quad (\mathcal{L} - q)J_{b^*}(x) + u(1 - J'_{b^*}(x)) \leq 0.$$

Let ℓ be an arbitrary admissible strategy and X^ℓ the associated controlled process with ruin time τ_0^ℓ . Hence, as $J_{b^*} \in \mathcal{C}^2(0, \infty)$, we can apply Itô's formula to obtain

$$\begin{aligned} & e^{-qt \wedge \tau_0^\ell} J_{b^*}(X_{t \wedge \tau_0^\ell}^\ell) - J_{b^*}(X_0^\ell) \\ &= M_t^\ell + \int_0^{t \wedge \tau_0^\ell} e^{-qs} ((\mathcal{L} - q)J_{b^*}(X_s^\ell) + (1 - J'_{b^*}(X_s^\ell))\ell_s^\ell - \ell_s^\ell) ds, \end{aligned}$$

where $M_t^\ell := \int_0^{t \wedge \tau_0^\ell} e^{-qs} \sigma(X_s^\ell) J'_{b^*}(X_s^\ell) dW_s$ is a local martingale.

Let $(\rho_n)_{n \geq 1}$ be a localizing sequence, and set $T_n := t \wedge \tau_0^\ell \wedge \rho_n$. Since ℓ is an admissible strategy, we can use (5.5) with $u = \ell_s$ to get

$$\mathbb{E}_x [e^{-qT_n} J_{b^*}(X_{T_n}^\ell)] - J_{b^*}(x) \leq -\mathbb{E}_x \left[\int_0^{T_n} e^{-qs} \ell_s ds \right].$$

Since $J_{b^*} \geq 0$, we can take limits as $t \rightarrow \infty$ and $n \rightarrow \infty$ and obtain

$$J_{b^*}(x) \geq J(x, \ell) = \mathbb{E}_x \left[\int_0^{\tau_0^\ell} e^{-qs} \ell_s ds \right] \quad \text{for all } x \geq 0.$$

Since this last inequality holds for any admissible strategy ℓ , then $J_{b^*} \geq V$.

Finally, let us prove that b^* is unique. First, we note that the above arguments and conclusion do not depend on the choice of b^* . Suppose there exists another $b^\dagger \in (0, \hat{b}]$ which is a root of (4.4). Then, we must have $J_{b^\dagger} = V$. It follows also that $J_{b^\dagger}(x) = \frac{\psi_q(x)}{\psi'_q(b^\dagger)}$ for all $x \in [0, b^\dagger]$. In particular, for each $x \in (0, b^* \wedge b^\dagger)$, we have

$$\frac{\psi_q(x)}{\psi'_q(b^\dagger)} = V(x) = \frac{\psi_q(x)}{\psi'_q(b^*)}.$$

As a consequence, we have $\psi'_q(b^\dagger) = \psi'_q(b^*)$ and, since ψ_q is strictly concave on $(0, \hat{b})$ by Lemma 2.4, we also have that $b^* = b^\dagger$. The proof is complete.

6. Examples. In this section, we first present some examples where the functions I_F , $\varphi_{q,F}$, and ψ_q are explicit, followed by a numerical analysis of a logistic-type model.

6.1. When μ and F are affine functions. Let μ and F be affine functions satisfying our standing assumptions and given by

$$\mu(x) = \mu_0 + \mu_1 x, \quad F(x) = F_0 + F_1 x, \quad \text{and} \quad c_i = \mu_i - F_i, \quad i = 0, 1.$$

From Proposition 3.2, we have

$$I_F(x) = \frac{1}{q} F_0 + \frac{F_1}{q(q - c_1)} (qx + c_0).$$

From Proposition 4.1, when $\frac{F_1}{(q - c_1)} - I_F(0)\varphi'_{q,F}(0) > 1$, there exists a positive solution b^* to

$$(6.1) \quad \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - \frac{\psi_q(b)}{\psi'_q(b)} = \frac{F_1}{q - c_1} \frac{\varphi_{q,F}(b)}{\varphi'_{q,F}(b)} - I_F(b),$$

and the value function is $V = J_{b^*}$; otherwise $V = J_0$.

To obtain expressions of the fundamental solutions $\varphi_{q,F}$ and ψ_q , we need to specify the form of the diffusion coefficient σ . The case when $\sigma(x) = \sigma > 0$ is a constant has been studied in [13, section 4.1]. As we see below, other specific choices of $\sigma(x)$ lead to semi-explicit expressions for $\varphi_{q,F}$ and ψ_q , in terms of well-known *special functions* such as the Kummer confluent hypergeometric function ${}_1F_1(a, b; z)$ and the confluent hypergeometric function of the second kind $U(a, b; z)$, which are linearly independent solutions of

$$zu'' + (b - z)u' - au = 0.$$

When $a, b > 0$, ${}_1F_1(a, b; z)$ can be represented as an integral

$$(6.2) \quad {}_1F_1(a, b; z) = \frac{\Gamma(b)}{\Gamma(b - a)\Gamma(a)} \int_0^1 e^{zt} t^{a-1} (1 - t)^{b-a-1} dt,$$

and when $a > 0$, $U(a, b; z)$ has the integral representation

$$(6.3) \quad U(a, b; z) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-zt} t^{a-1} (1 + t)^{b-a-1} dt.$$

We refer the reader to [1] for more details about these functions, and to Appendix C for the proofs of what follows.

6.1.1. When σ is an affine function. Assume $\sigma(x) = \sigma_0 + \sigma_1 x$, with $\sigma_0, \sigma_1 > 0$. For example, if $\mu_0 = F_0 = \sigma_0 = 0$, then X and Y are geometric Brownian motions.

Let $\Delta_F > 0$ with $\Delta_F^2 = (\sigma_1^2 - 2c_1)^2 + 8q\sigma_1^2$, $a_F = (\Delta_F - \sigma_1^2 + 2c_1)/(2\sigma_1^2)$, and $b_F = 1 + \Delta_F/\sigma_1^2$. Then,

$$(6.4) \quad \varphi_{q,F}(x) = \left(1 + \frac{\sigma_1}{\sigma_0} x\right)^{-a_F} \times \frac{{}_1F_1\left(a_F, b_F; \frac{2(c_0\sigma_1 - c_1\sigma_0)}{\sigma_1^2(\sigma_0 + \sigma_1 x)}\right)}{{}_1F_1\left(a_F, b_F; \frac{2(c_0\sigma_1 - c_1\sigma_0)}{\sigma_1^2\sigma_0}\right)}.$$

We easily observe that $\varphi_{q,F}(0) = 1$ and $\lim_{x \rightarrow +\infty} \varphi_{q,F}(x) = 0$. We now introduce

$$(6.5) \quad \psi_q(x) = \kappa \left(1 + \frac{\sigma_1}{\sigma_0} x\right)^{-a_0 + b_0 - 1} \times \frac{{}_1F_1\left(1 + a_0 - b_0, 2 - b_0; \frac{2(c_0\sigma_1 - c_1\sigma_0)}{\sigma_1^2(\sigma_0 + \sigma_1 x)}\right)}{{}_1F_1\left(1 + a_0 - b_0, 2 - b_0; \frac{2(c_0\sigma_1 - c_1\sigma_0)}{\sigma_1^2\sigma_0}\right)} - \kappa \varphi_0^q(x),$$

where κ is explicitly determined by $(\psi_q)'(0) = 1$, and $a_0 := a_F$, $b_0 := b_F$, and $\varphi_0^q := \varphi_{q,F}$ for $F \equiv 0$.

6.1.2. When σ^2 is an affine function. Assume $\sigma(x) = \sqrt{\sigma_0 + \sigma_1 x}$, with $\sigma_0, \sigma_1 > 0$. When $\sigma_0 = 0$, X and Y are square-root or Cox–Ingersoll–Ross processes.

Let $a_F = \frac{q}{|c_1|}$ and $b_F = 1 + 2\frac{c_1\sigma_0 - c_0\sigma_1}{\sigma_1^2}$. Then,

$$(6.6) \quad \varphi_{q,F}(x) = \begin{cases} e^{-2\frac{c_1}{\sigma_1}x} \left(1 + \frac{\sigma_1}{\sigma_0}x\right)^{b_F} \times \frac{U\left(1+a_F, 1+b_F; \frac{2c_1(\sigma_0+\sigma_1 x)}{\sigma_1^2}\right)}{U\left(1+a_F, 1+b_F; \frac{2c_1\sigma_0}{\sigma_1^2}\right)} & \text{when } c_1 > 0, \\ \frac{U\left(a_F, 1-b_F; -\frac{2c_1(\sigma_0+\sigma_1 x)}{\sigma_1^2}\right)}{U\left(a_F, 1-b_F; -\frac{2c_1\sigma_0}{\sigma_1^2}\right)} & \text{when } c_1 < 0. \end{cases}$$

Using (6.3), we easily observe that $\varphi_{q,F}(0) = 1$ and $\lim_{x \rightarrow +\infty} \varphi_{q,F}(x) = 0$. Moreover,

$$\psi_q(x) = \begin{cases} \kappa e^{-2\frac{\mu_1}{\sigma_1}x} \left(1 + \frac{\sigma_1}{\sigma_0}x\right)^{b_0} \times \frac{{}_1F_1\left(1+a_0, 1+b_0; \frac{2\mu_1(\sigma_0+\sigma_1 x)}{\sigma_1^2}\right)}{{}_1F_1\left(1+a_0, 1+b_0; \frac{2\mu_1\sigma_0}{\sigma_1^2}\right)} - \kappa\varphi_0^q(x) & \text{when } \mu_1 > 0, \\ \kappa \left(1 + \frac{\sigma_1}{\sigma_0}x\right)^{b_0} \times \frac{{}_1F_1\left(a_0+b_0, b_0; -\frac{2\mu_1(\sigma_0+\sigma_1 x)}{\sigma_1^2}\right)}{{}_1F_1\left(a_0+b_0, b_0; -\frac{2\mu_1\sigma_0}{\sigma_1^2}\right)} - \kappa\varphi_0^q(x) & \text{when } \mu_1 < 0, \end{cases}$$

where κ is explicitly determined by $(\psi_q)'(0) = 1$, and $a_0 := a_F$, $b_0 := b_F$, and $\varphi_0^q := \varphi_{q,F}$ for $F \equiv 0$. When $c_1 = 0$ or $\mu_1 = 0$, the functions $\varphi_{q,F}$ and ψ_q also have an explicit expression using modified Bessel functions.

In Figure 1, the curves of $\varphi_{q,0}$ and ψ_q are drawn in the above cases for a given set of parameters, including the case σ constant (see [13, section 4.1]). We observe the properties announced in Lemma 2.4, i.e., when $\mu_1 < q$, $\varphi_{q,0}$ is strictly convex and ψ_q is concave-convex.

We now compute b^* numerically, as the root of (6.1), and the value function J_{b^*} , given by (4.5), for different values of (F_0, F_1) . In Figure 2, we observe the dependence of b^* on $(F_0, F_1) \in \mathbb{R}_+^2$, the other parameters being fixed. The white area below the red curve represents the case $b^* = 0$, and the area above the red curve coincides with the case $b^* > 0$: the darker the blue, the higher the value of b^* . Using (4.5), the curves of J_{b^*} and their corresponding first derivatives are drawn in Figure 3 for different values of (σ_0, σ_1) . In all cases, we observe that the first derivative is decreasing which coincides with the fact that J_{b^*} is concave on $\mathbb{R}_+$.

6.2. A logistic-type model. We consider a modification of the logistic model, where the drift coefficient is taken as $\mu(x) = \mu_0 + \mu_1 x(1 - x/K)$ and the diffusion coefficient is given by $\sigma(x) = \sqrt{\sigma_0 + \sigma_1 x}$. In the literature of logistic models, the parameter μ_1 is known as the *population growth rate*, while the parameter K is known as the *carrying capacity of the environment* (see, e.g., [3, section 2]). Note that the parameters $\mu_0 > 0$ and $\sigma_0 > 0$ are important in order to satisfy the assumptions in Hypothesis 2.1. We also remark that, given a discount factor $q > 0$, the value of μ_1 is chosen so that $\mu_1 \leq q$, as one can easily verify that $\mu'(0) = \mu_1$. Throughout this section, we will consider that the bound function is of the form $F(x) = F_0 + F_1 x$ for all $x \in \mathbb{R}$.

While in this case we do not have semi-explicit expressions for the functions ψ_q , $\varphi_{q,F}$, and I_F , we are still able to perform computations by solving numerically their associated ODEs using the finite differences scheme; this allows us to find numerically $\hat{b}$, the unique inflection point of ψ_q' . Consequently, we can find a numerical solution to (4.4).

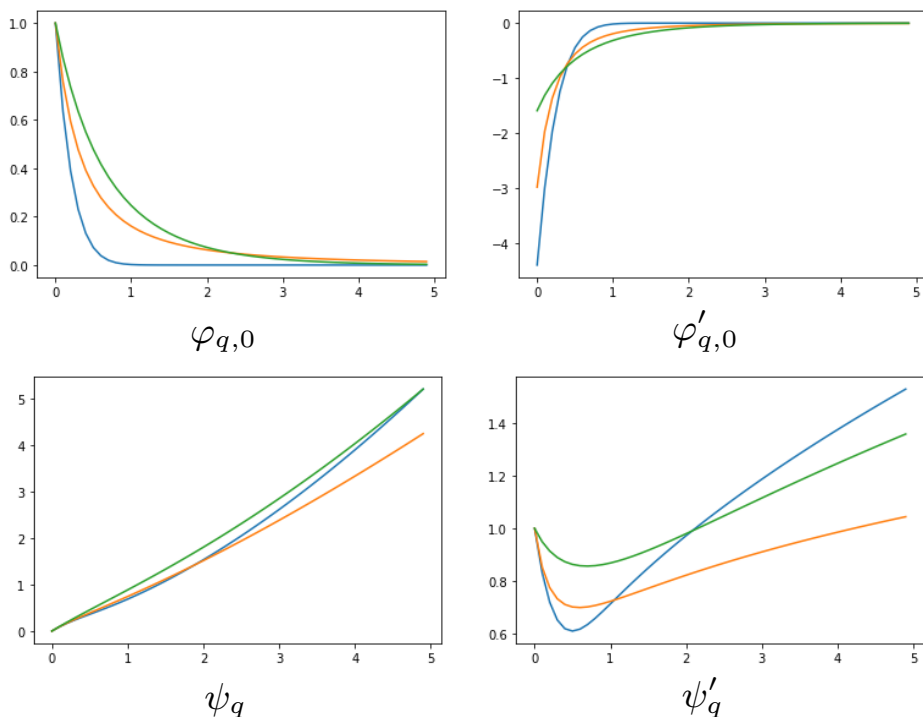


FIG. 1. The functions $\varphi_{q,0}$ and ψ_q and their first derivatives when $\mu_0 = 0.09$, $\mu_1 = 0.21$, $\sigma_0 = 0.3$, $\sigma_1 = 0.5$, and $q = 0.33$: In blue when $\sigma(x) = \sigma_0$, in orange when $\sigma(x) = \sigma_0 + \sigma_1 x$, and in green when $\sigma^2(x) = \sigma_0 + \sigma_1 x$.

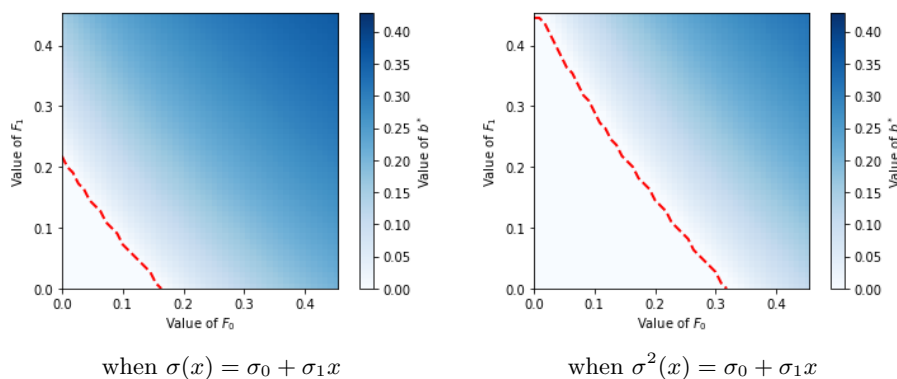


FIG. 2. Heat maps of b^* as a function of $(F_0, F_1) \in \mathbb{R}_+^2$, when $\mu_0 = 0.09$, $\mu_1 = 0.21$, $\sigma_0 = 0.3$, $\sigma_1 = 0.5$, and $q = 0.33$.

In Figure 4, we illustrate the sensitivity of b^* for different values of (F_0, F_1) . In particular, the region below the red curve represents the values of (F_0, F_1) for which $b^* = 0$; whereas the region above the red curve shows the values where $b^* > 0$. We remark that the darker the shade, the larger the value of b^* . We can observe that, for a fixed value of F_0 , b^* is nondecreasing with respect to F_1 , and, likewise for a fixed value of F_1 , then b^* is nondecreasing with respect to F_0 .

In Figure 5, we illustrate the effect of the diffusion coefficient on the value function, as well as in the optimal refraction barrier. To this end, first we fixed a value of σ_1

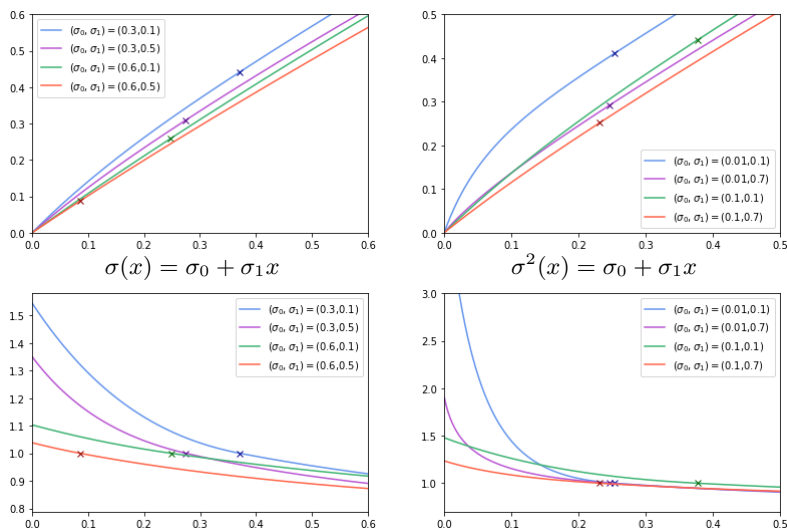


FIG. 3. The value function J_{b^*} (on the first row) and its first derivative (on the second row) when $\mu_0 = 0.09$, $\mu_1 = 0.21$, $F_0 = 0.3$, $F_1 = 0.3$, and $q = 0.33$, and for different couples (σ_0, σ_1) . The optimal refractive threshold is marked with a cross.

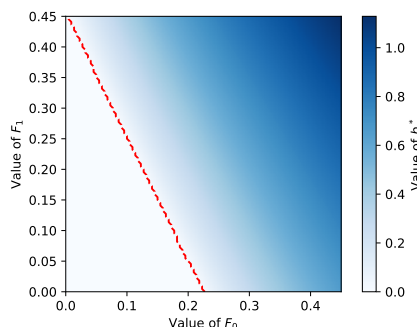


FIG. 4. Heat map of b^* as a function of $(F_0, F_1) \in \mathbb{R}_+^2$, when $\mu_0 = 0.15$, $\mu_1 = 0.21$, $K = 10$, $\sigma_0 = 0.75$, $\sigma_1 = 0.5$, and $q = 0.33$.

and proceeded to solve the optimization problem for different values of σ_0 ; afterwards, we fixed a value of σ_0 and solved the control problem for different values of σ_1 . First, we confirm the concavity of the value function on all cases. Regarding the effect of σ_0 and σ_1 , we notice that if $\sigma_0 \leq \tilde{\sigma}_0$ while σ_1 is fixed, then the corresponding value functions J_{b^*} and $J_{\tilde{b}^*}$ satisfy $J_{\tilde{b}^*} \leq J_{b^*}$. We observe a similar effect with respect to σ_1 while σ_0 is fixed.

Appendix A. Finiteness and transversality property.

Proof of Proposition 3.1. Let $h(x) = 2(1 - |x| \wedge 1)$ for $x \in \mathbb{R}$. Note that $h(x) \in [0, 2]$ and $h(x) = 0$ whenever $|x| > 1$. Define f by $f''(x) = h(x)$ such that $f(-1) = 1$ and $f'(-1) = -1$. Then, $f(x) \geq |x|$ and $f'(x) = \text{sgn}(x)$ on $\mathbb{R} \setminus [-1, 1]$. Define the constants $\tilde{q} := \max(\mu'(0+), q/2)$, $\tilde{\sigma} := \sup_{|x| < 1} \sigma(x)$, and

$$C = \max \left\{ \sup_{|x| < 1} [-\tilde{q}f(x) + f'(x)(\mu(x) - F(x))], 0 \right\}.$$

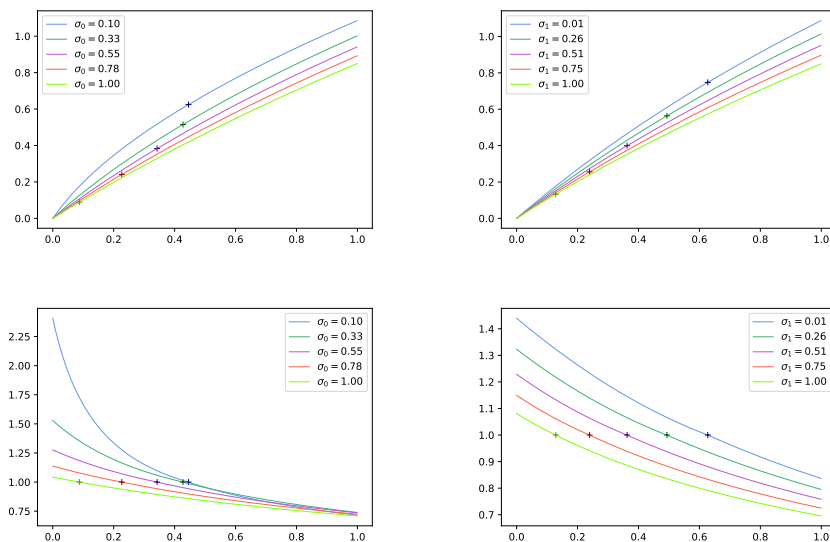


FIG. 5. Effect of the values of σ_0 and σ_1 when $q = 0.33$, $\mu_0 = 0.25$, $\mu_1 = 0.3$, $K = 5$. The bound function is $F(x) = 0.15 + 0.25x$. The optimal refraction threshold is marked as a cross. Left: Value function J_b^* (on the top row) and its first derivative (on the second row) for different values of σ_0 when $\sigma_1 = 0.75$ is fixed. Right: Value function J_b^* (on the top row) and its first derivative (on the second row) for different values of σ_1 when $\sigma_0 = 0.75$ is fixed.

Note that $\tilde{q} > 0$ and $\mu'(0+) \leq \tilde{q} < q$. Because f is smooth enough and f'' is bounded, we can apply Itô's formula in order to obtain

$$\begin{aligned} \mathbb{E}_x [e^{-\tilde{q}t} f(X_t)] - f(x) &= \mathbb{E}_x \left[\int_0^t e^{-\tilde{q}s} \left(-\tilde{q}f(X_s) + f'(X_s)(\mu(X_s) - F(X_s)) + \frac{1}{2}\sigma^2(X_s)f''(X_s) \right) ds \right] \\ &= \mathbb{E}_x \left[\int_0^t e^{-\tilde{q}s} (\mathbf{1}_{|X_s| > 1} (-\tilde{q}|X_s| + \text{sgn}(X_s)(\mu(X_s) - F(X_s))) + C\mathbf{1}_{|X_s| \leq 1} + \bar{\sigma}^2) ds \right]. \end{aligned}$$

Observe that the term $\mathbb{E}_x[\int_0^t e^{-\tilde{q}s} (C\mathbf{1}_{\{|X_s| \leq 1\}} + \bar{\sigma}^2) ds]$ is nonnegative, and we have the upper bound

$$(A.1) \quad \mathbb{E}_x \left[\int_0^t e^{-\tilde{q}s} (C\mathbf{1}_{\{|X_s| \leq 1\}} + \bar{\sigma}^2) ds \right] \leq \frac{C + \bar{\sigma}^2}{\tilde{q}}.$$

It remains to study $\mathbb{E}_x[\int_0^t e^{-\tilde{q}s} \mathbf{1}_{\{|X_s| > 1\}} (-\tilde{q}|X_s| + \text{sgn}(X_s)(\mu(X_s) - F(X_s))) ds]$. To this end, on one hand, if $z > 1$, we have that

$$(A.2) \quad -\tilde{q}z + (\mu(z) - F(z)) \leq -\tilde{q}z + \mu(z) \leq \mu(0),$$

where in the first inequality we used that F is nonnegative on $(0, \infty)$, and in the second inequality we used that μ is concave, hence $\mu(z) \leq \mu(0) + \mu'(0+)z$, together with the fact that $\tilde{q} \geq \mu'(0+)$.

On the other hand, if $z < -1$, we have

$$(A.3) \quad \tilde{q}z - (\mu(z) - F(z)) = (\tilde{q} - \mu'(0+) + F'(0+))z + F(0) - \mu(0) \leq F(0) - \mu(0) \leq F(0),$$

where in the equality we simply substituted the value of the extensions of μ and F , in the first inequality we used that $\tilde{q} \geq \mu'(0+)$ and $F'(0+)z \leq 0$ because $F'(0+) \geq 0$, and in the second inequality we used that $\mu(0) > 0$.

Using the upper bounds (A.2) and (A.3), we have that

$$(A.4) \quad \mathbb{E}_x \left[\int_0^t e^{-\tilde{q}s} \mathbf{1}_{\{|X_s|>1\}} (-\tilde{q}|X_s| + \operatorname{sgn}(X_s)(\mu(X_s) - F(X_s))) ds \right] \\ \leq \mathbb{E}_x \left[\int_0^t e^{-\tilde{q}s} \mathbf{1}_{\{|X_s|>1\}} \mu(0) \vee F(0) ds \right] \leq \frac{\mu(0) \vee F(0)}{\tilde{q}}.$$

Hence, by combining (A.1) and (A.4) we have that

$$\mathbb{E}_x [e^{-\tilde{q}t} f(X_t)] \leq f(x) + \frac{\mu(0) \vee F(0) + C + \bar{\sigma}^2}{\tilde{q}}.$$

Let $H_x := f(x) + \frac{\mu(0) \vee F(0) + C + \bar{\sigma}^2}{\tilde{q}}$. Then, since $f(x) \geq |x|$, we get the upper bound

$$(A.5) \quad \mathbb{E}_x [e^{-\tilde{q}t} |X_t|] \leq \mathbb{E}_x [e^{-\tilde{q}t} f(X_t)] \leq H_x.$$

Finally, using that F is concave and (A.5), we obtain for $q > \mu'(0+)$

$$\begin{aligned} \mathbb{E}_x \left[\int_0^\infty e^{-qt} |F(X_t)| dt \right] &\leq \mathbb{E}_x \left[\int_0^\infty e^{-qt} (F(0) + F'(0)|X_t|) dt \right] \\ &\leq \frac{F(0)}{q} + F'(0) \int_0^\infty e^{-(q-\tilde{q})t} \mathbb{E}_x [e^{-\tilde{q}t} |X_t|] dt \\ &\leq \frac{F(0)}{q} + F'(0) \frac{H_x}{q-\tilde{q}} < \infty. \end{aligned} \quad \square$$

LEMMA A.1. Under Hypothesis 2.1, $\mathbb{E}_x[e^{-qT} X_T] \rightarrow 0$ as $T \rightarrow \infty$.

Proof. Denoting $\tilde{q} = \mu'(0+)$, it is proved in the proof of Proposition 3.1 that

$$\mathbb{E}_x [e^{-qT} |X_T|] = e^{-(q-\tilde{q})T} \mathbb{E}_x [e^{-\tilde{q}T} |X_T|] \leq e^{-(q-\tilde{q})T} H_x,$$

where H_x is defined above. The result follows since H_x is independent of T . $\square$

Appendix B. Properties of the concave envelope. The concave envelope U of a real-valued function I is defined as

$$(B.1) \quad U(z) := \inf \{ \Phi(z) : \Phi \text{ is concave and } \Phi \geq I \}.$$

The function U takes its values in $\mathbb{R} \cup \{+\infty\}$ and is itself a concave function. We refer the reader to [8, Chap. B, sect. 2.5] and [17] for a broader and more detailed exposition of *convex envelopes* and their properties. The proof of the following lemma is similar to that of [14, Lemma 2.2] and is omitted.

LEMMA B.1. Let I be a continuous function on $\mathbb{R}$ and U its concave envelope. Assume that U takes finite values. If there exists $(a, b) \subset \mathbb{R}$ such that $I < U$ on (a, b) , then U is affine on (a, b) .

LEMMA B.2. Let I be a function in $\mathcal{C}^2(\mathbb{R})$ and U its concave envelope. Let $x_0 \in \mathbb{R}$ such that $I(x_0) < U(x_0) < \infty$.

1. If there exists $\hat{x} \in (-\infty, x_0)$ such that $U(\hat{x}) = I(\hat{x})$ and $I < U$ on $(\hat{x}, x_0)$, then U is differentiable at $\hat{x}$ and

$$U(\hat{x}) = I(\hat{x}), \quad U'(\hat{x}) = I'(\hat{x}), \quad \text{and} \quad I''(\hat{x}) \leq 0.$$

2. If there exists $\hat{y} \in (x_0, +\infty)$ such that $U(\hat{y}) = I(\hat{y})$ and $I < U$ on $(x_0, \hat{y})$, then U is differentiable at $\hat{y}$ and

$$U(\hat{y}) = I(\hat{y}), \quad U'(\hat{y}) = I'(\hat{y}), \quad \text{and} \quad I''(\hat{y}) \leq 0.$$

Proof. Assume case 1 is satisfied. From Lemma B.1, we know that U is affine on $[\hat{x}, x_0]$. We denote by L the affine function

$$L(x) = U(\hat{x}) + U'(\hat{x}+)(x - \hat{x}) = I(\hat{x}) + U'(\hat{x}+)(x - \hat{x}),$$

where $U'(\hat{x}+) = \lim_{x \rightarrow \hat{x}+} U'(x)$ is the right derivative of U at $\hat{x}$.

We have $L = U$ on $[\hat{x}, x_0]$ and $U \leq L$ on $\mathbb{R}$ by concavity of U . If $I'(\hat{x}) > U'(\hat{x}+)$, then there is $x > \hat{x}$ such that $I(x) > U(x)$, which is impossible since $U \geq I$ by definition. If $I'(\hat{x}) < U'(\hat{x}+)$, then there is $x < \hat{x}$ such that $I(x) > L(x)$, which is also impossible since $U \leq L$. Then $U'(\hat{x}+) = I'(\hat{x})$.

Since $I \in \mathcal{C}^2(\mathbb{R})$, by Taylor's theorem, there exists ε such that, for $x \geq \hat{x}$,

$$\begin{aligned} I(x) &= I(\hat{x}) + I'(\hat{x})(x - \hat{x}) + \frac{I''(\hat{x})}{2}(x - \hat{x})^2 + \varepsilon(x)(x - \hat{x})^2 \\ &= U(x) + \frac{I''(\hat{x})}{2}(x - \hat{x})^2 + \varepsilon(x)(x - \hat{x})^2, \end{aligned}$$

with $\lim_{x \rightarrow \hat{x}} \varepsilon(x) = 0$. As $U \geq I$, we deduce that $I''(\hat{x}) \leq 0$. Finally, let us note that on a small interval of the form $[\tilde{x}, \hat{x}]$, with $\tilde{x} < \hat{x}$, we have either $U(x) = I(x)$, and then U is differentiable in $\hat{x}$, with $U'(\hat{x}) = I'(\hat{x})$, or $U(x) > I(x)$, and then by the previous proof $U'(\hat{x}-) = I'(\hat{x})$. The proof of case 2 is similar. $\square$

Appendix C. Fundamental solutions to the ODE for specific models. We prove in this section the expressions of the fundamental solutions given in section 6.1, when μ and F are affine functions.

When σ is affine. The proof scheme is the same as in the previous case. By [1, Chapter 13], two linearly independent solutions of the following ODE

$$(\mathbf{E}(a, b)) \quad zy'' + (b - z)y' - ay = 0.$$

are the Kummer confluent Hypergeometric function ${}_1F_1(a, b; z)$ of the first kind, and the confluent hypergeometric function of the second kind $U(a, b; z)$, where $U(a, b; z)$ is indeed an explicit linear combination of ${}_1F_1(a, b; z)$ and $z^{1-b}{}_1F_1(1 + a - b, 2 - b; z)$ (see [1, eq. 13.1.3]).

We will prove that $u(x) = z^a y(\alpha z)$, with $z = (\sigma_0 + \sigma_1 x)^{-1}$, $a = (\Delta - \sigma_1^2 + 2c_1)/(2\sigma_1^2)$, $b = 1 + \Delta/\sigma_1^2$, $\Delta^2 = (\sigma_1^2 - 2c_1)^2 + 8q\sigma_1^2$, $\alpha = \frac{2(c_0\sigma_1 - c_1\sigma_0)}{\sigma_1^2}$, and y a solution of $\mathbf{E}(a, b)$, is a solution of

$$(C.1) \quad (\mathcal{L}_F - q)u = \frac{1}{2}(\sigma_0 + \sigma_1 x)^2 u'' + (c_0 + c_1 x)u' - qu = 0.$$

As $a, b > 0$, from the integral representation (6.2) of ${}_1F_1$, we deduce that $\varphi_{q,F}$ defined by (6.4) satisfies $\varphi_{q,F}(0) = 1$ and $\lim_{x \rightarrow +\infty} \varphi_{q,F}(x) = 0$. The function ψ_q is thus a linear combination of $\varphi_{q,0}$ and $z^a U(a, b; \alpha z)$ where the coefficients are determined by the boundary conditions of the second fundamental solutions. In fact, we used $z^{1-b}{}_1F_1(1 + a - b, 2 - b; z)$ instead of U as second solution of $\mathbf{E}(a, b)$ in expression (6.5) of section 6.1.

Let us now prove that $u(x) = z^a y(\alpha z)$, with z , a , b , and α defined above, is a solution of the ODE (C.1). We compute the first and second derivative of u with respect to x :

$$\begin{aligned} u'(x) &= -a\sigma_1 z^{a+1} y(\alpha z) - \sigma_1 \alpha z^{a+2} y'(\alpha z) \\ u''(x) &= a(a+1)\sigma_1^2 z^{a+2} y(\alpha z) + 2(a+1)\sigma_1^2 \alpha z^{a+3} y'(\alpha z) + \sigma_1^2 \alpha^2 z^{a+4} y''(\alpha z). \end{aligned}$$

Consequently,

$$\begin{aligned} & \frac{1}{2}(\sigma_0 + \sigma_1 x)^2 u''(x) + (c_0 + c_1 x)u'(x) - qu(x) \\ &= \frac{1}{2z^2} u''(x) + (c_0 + c_1 x)u'(x) - qu(x) \\ &= \left(\frac{a(a+1)}{2} \sigma_1^2 - a\sigma_1 \frac{c_0 + c_1 x}{\sigma_0 + \sigma_1 x} - q \right) u(x) \\ & \quad + [-(c_0 + c_1 x) + (a+1)\sigma_1(\sigma_0 + \sigma_1 x)] \sigma_1 \alpha z^{a+2} y'(\alpha z) + \frac{\sigma_1^2 \alpha^2}{2} z^{a+2} y''(\alpha z). \end{aligned}$$

By definition of the parameter a , we have $\frac{a(a+1)}{2} \sigma_1^2 - a\sigma_1 - q = 0$. Consequently,

$$\begin{aligned} & \frac{a(a+1)}{2} \sigma_1^2 - a\sigma_1 \frac{c_0 + c_1 x}{\sigma_0 + \sigma_1 x} - q \\ &= ac_1 - a\sigma_1 \frac{c_0 + c_1 x}{\sigma_0 + \sigma_1 x} = -a\alpha \frac{\sigma_1^2}{2} z, \\ & -(c_0 + c_1 x) + (a+1)\sigma_1(\sigma_0 + \sigma_1 x) = \frac{\sigma_1}{2} b(\sigma_0 + \sigma_1 x) - \frac{\sigma_1}{2} \alpha = \frac{\sigma_1}{2z} (b - \alpha z). \end{aligned}$$

We finally obtain that u is a solution of the ODE (C.1) since y is solution of $\mathbf{E}(a, b)$.

When σ^2 is affine. We only study the case $c_1 > 0$, the proof for $c_1 < 0$ being similar. We will prove that $u(x) = e^{-2\frac{c_1}{\sigma_1}x} z^b y(\alpha z)$ with $a = \frac{q}{c_1}$, $b = 1 + 2\frac{c_1\sigma_0 - c_0\sigma_1}{\sigma_1^2}$, $\alpha = 2\frac{c_1}{\sigma_1^2}$, $z = \sigma_0 + \sigma_1 x$, and y solution of the differential equation $\mathbf{E}(1+a, 1+b)$, is a solution of

$$(C.2) \quad (\mathcal{L}_F - q)u = \frac{1}{2}(\sigma_0 + \sigma_1 x)u'' + (c_0 + c_1 x)u' - qu = 0.$$

As $a > 0$, from the integral form (6.3) of U , we deduce that $\varphi_{q,F}$ defined by (6.6) is solution of the ODE (C.2) and satisfies $\varphi_{q,F}(0) = 1$ with $\lim_{x \rightarrow +\infty} \varphi_{q,F}(x) = 0$. The expression of the second fundamental solution ψ_q follows.

We now prove that $u(x) = e^{-2\frac{c_1}{\sigma_1}x} z^b y(\alpha z)$, is a solution of (C.2). First,

$$\begin{aligned} & \frac{1}{2}(\sigma_0 + \sigma_1 x)u'' + (c_0 + c_1 x)u' - qu \\ &= \left(2\frac{c_1^2}{\sigma_1^2} z - 2c_1 b + \frac{b(b-1)}{2z} \sigma_1^2 + \left(-2\frac{c_1}{\sigma_1} + \frac{b\sigma_1}{z} \right) (c_0 + c_1 x) - q \right) u(x) \\ & \quad + \alpha [-2c_1 z + b\sigma_1^2 + \sigma_1(c_0 + c_1 x)] e^{-2\frac{c_1}{\sigma_1}x} z^b y'(\alpha z) + \frac{\alpha^2 \sigma_1^2}{2} e^{-2\frac{c_1}{\sigma_1}x} z^{b+1} y''(\alpha z). \end{aligned}$$

Then, as $x = \frac{z - \sigma_0}{\sigma_1}$, we observe that

$$\begin{aligned} & 2\frac{c_1^2}{\sigma_1^2} z - 2c_1 b + \frac{b(b-1)}{2z} \sigma_1^2 + \left(-2\frac{c_1}{\sigma_1} + \frac{b\sigma_1}{z} \right) (c_0 + c_1 x) = -c_1, \\ & -2c_1 z + b\sigma_1^2 + \sigma_1(c_0 + c_1 x) = -c_1 z + \sigma_1^2 + (c_1 \sigma_0 - c_0 \sigma_1) = \frac{\sigma_1^2}{2} (1 + b - \alpha z). \end{aligned}$$

Thus, u is solution of the ODE (C.2) since y is solution of $\mathbf{E}(1+a, 1+b)$.

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